

MIDAS GEN NX API — Plug-in Application Cases

GEN NX API CHALLENGE 2026 | 20 Plug-ins

A plug-in that automatically models walls and perimeter lintel beams from structural drawings. It automatically splits wall intersections and aligns/merges nearby nodes created by minor coordinate discrepancies in the drawing, quickly producing an accurate analysis model free of disconnected elements and unnecessary nodes.

Step-by-step guidance for first-time users An intuitive status bar at the top and pulsing highlights on input fields show the current progress and the next action, so even first-time users can pick up the full workflow quickly.

Story-by-story DXF wall extraction and analysis Automatically extracts and analyzes wall centerlines and wall information from each story's DXF drawing to build the set of modeling targets.

Automatic wall and lintel beam modeling Automatically models each story's walls and opening lintel beams based on the analyzed drawing.

Automatic splitting at intersections Automatically detects wall intersections and splits elements there, producing a well-connected analysis model with shared nodes.

Automatic merging of nearby nodes Prevents multiple nodes from being created at the same joint due to minor coordinate discrepancies in the drawing, automatically aligning and merging nearby nodes within a configurable tolerance.

WallMark Auto Mapper

WallMark Auto Mapper is a plug-in that extracts wall names and location information from a DXF structural plan, automatically matches them against the wall locations already assigned to each Wall ID in the MIDAS GEN NX analysis model, and assigns the drawing's wall names to the matching Wall ID as its Wall Mark. The automatic matching results can be reviewed and edited in a preview and result table before being

selectively applied to the model.

Extract wall names and locations from the DXF drawing Extracts wall names and location information from a specified layer of the selected DXF structural plan, automatically building the drawing-side dataset to be matched against the GEN NX model.

Compute wall centroid coordinates per GEN NX Wall ID Selects the wall elements corresponding to the chosen story and Wall Type (Membrane or Plate), and uses their Node and Element data to compute the wall centroid coordinates for each existing Wall ID.

Automatic matching of drawing wall data to Wall ID Compares the wall names and locations extracted from the DXF structural plan against the wall centroid coordinates of each Wall ID in the GEN NX analysis model, within a tolerance distance, and links the closest candidate. Matching results are classified as matched, needs review, or failed, and shown in the preview and result table.

Review, apply, and manage Wall Marks After reviewing and editing the matching results between drawing wall names and Wall IDs in the preview and result table, assigns the selected drawing wall name as the Wall Mark to one or more matching Wall IDs. Supports adding, editing, and deleting existing Wall Marks, Excel import/export, and PDF result table output.

Vibration Analysis Assistant is a plug-in that lets you input, review, and apply the Time History Function, Time History Load Case, and Dynamic Nodal Load data needed to check pedestrian-induced vibration of slabs or beam members in a MIDAS GEN NX analysis model — all from a single screen. Graphs and tables let you check load characteristics directly, and the GEN NX API lets you push the Time History data to the model quickly.

Batch input for pedestrian loads Brings together the function, analysis case, damping condition, and nodal load inputs needed for pedestrian-load review, all configurable from one screen.

Real-time review of load functions As soon as an input changes, the graph and table for the IABSE pedestrian load time function update in sync, so you can immediately check load magnitude, duration, and time step.

Automatic application of GEN NX API input data Creates, edits, and deletes input data such as the Time History Function, Time History Load Case, and Dynamic Nodal Load through the API, applying them to the GEN NX model right away.

Fewer repetitive steps and input errors Cuts down on the manual work of moving between scattered feature windows, letting you create vibration-review conditions to a consistent standard, quickly and reliably.

A plug-in that automatically computes the seismic-load Cm Factor (correction factor) needed when generating load combinations in MIDAS GEN NX.

Compute the seismic load correction factor Automatically calculates the seismic-load Cm Factor (correction factor) needed when generating load combinations in MIDAS GEN NX.

Automatic input Pulls in, in one step, all the values already entered in the model that are needed to derive the correction factor.

Applies KDS 41 17 00 §4.2.2 Automatically judges and applies both of the following: (2) where bedrock depth exceeds 20 m and the average shear-wave velocity of the soil is 360 m/s or more, apply 80% of F_v as specified in Table 4.2-2; (3) where the soil classification is S5 and the bedrock depth is unclear, apply 110% of F_a and F_v as specified in Tables 4.2-1 and 4.2-2.

Output a calculation report for the correction factor Outputs a calculation report documenting exactly what data the correction factor was derived from.

WallStiffnessAuto reads RC wall design (KDS-41-20-2022) results directly through the GEN NX API, and progressively lowers the Wall Stiffness Scale Factor (WSSF) of any wall that exceeds the code check, on its own, until all NG results are cleared.

Automatic detection of NG members After running the structural analysis and wall design, automatically extracts the walls that exceed the code check by story/Wall ID and shows the governing stress ratio alongside each one.

Configure reduction conditions Set the reduction step size, the lower bound for the factor, and the maximum number of iterations, and the automatic loop runs within that range.

Automatic iterative reduction Each iteration applies the reduced stiffness, re-runs the analysis, then re-checks the design; the loop stops automatically once all NG results are cleared or the factor reaches its lower bound.

Restore the factor At any point during the iterations, the reduced factor can be restored to its pre-reduction value (1.0) with a single button click.

A plug-in that, from a single screen, splits rectangular walls in GEN NX at a reference length, cleans up and merges them to fit the surrounding node grid, and automatically corrects them based on their layout direction.

Wall review Because every wall is processed with the same rule, model consistency improves regardless of who built it, and the model is easy to revise during review and editing.

A plug-in that automatically computes the response-spectrum scale-up factor C_m required by the seismic design code and automatically generates the corresponding design load combinations in GEN NX. It automatically pulls the analysis results and load-case information needed to derive C_m straight from GEN NX, so the entire workflow — from checking analysis results, through deriving C_m , to generating design load combinations and outputting a report — runs as one continuous process without

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Automatic computation of the scale-up factor C_m Compares the base shear from the equivalent static analysis and the response spectrum analysis by direction, and automatically computes the response-spectrum scale-up factor C_m required by the seismic design code.

Automatic linkage of GEN NX analysis data Automatically pulls the analysis results and load-case information needed to derive C_m directly from GEN NX, so you can compute it conveniently without repeated result checking or manual data entry.

Generate load combinations and output a report Automatically generates the design load combinations that apply the computed C_m in GEN NX, and provides a report-output feature summarizing the C_m calculation process and its basis.

This plug-in is a practical tool that automates the main steps of working with single-pole and double-pole scaffolding in MIDAS GEN NX — from building the model, through applying loads, running the structural analysis and steel code check, saving result images, and producing the scaffold structural review report.

Automatic scaffold model generation Automatically generates single-pole or double-pole scaffold models based on the entered conditions.

Automatic application of loads and analysis conditions Automatically applies wind load, noise-barrier and debris-net/working-platform loads, boundary conditions, and load combinations.

Structural analysis and steel code check Runs the structural analysis and steel code check in sequence, and shows the review result as OK or NG.

Save analysis result images and produce the review report Automatically saves analysis result images and generates a scaffold structural review report incorporating the input data and result images.

A plug-in that uses MIDAS GEN NX's RC design API (DCRM, CD/BD-ANAL, design result tables) to automatically sweep candidate bar sizes for the main reinforcement of RC columns and beams, derive the minimum steel arrangement per member that satisfies the check ratio (CHK), and unify members sharing the same section (same column mark or beam mark) into a single standard rebar arrangement.

Result-driven design Uses MIDAS GEN NX's RC design API (DCRM, CD/BD-ANAL, design result tables) to automatically sweep candidate bar sizes for the main reinforcement of RC columns and beams.

Member review Derives, per member, the minimum steel arrangement that satisfies the check ratio (CHK), and unifies members sharing the same section (same column mark or beam mark) into a single standard rebar arrangement.

Member design Every standard rebar arrangement is re-verified by re-running the GEN NX design, guaranteeing safety, and a governing-factor diagnosis shows which members can still be reduced further.

Result review Outputs the rebar schedule and comparison results as PDF immediately, cutting the time needed to prepare design, quantity, and review documents.

A plug-in that reads an RC beam's section, material, rebar arrangement, and the analysis moment for each load case from a MIDAS GEN NX model via the API, computes short-term and long-term deflection automatically using the effective section stiffness (I_e, Branson) per KDS 14 20 30 §4.2, compares it against the allowable deflection, and outputs an A4 structural calculation sheet (deflection review report).

Automatic model data retrieval Automatically extracts the RC beam's section, material, rebar arrangement, and analysis moment via the API.

Automatic deflection calculation Automatically derives short-term and long-term deflection per the KDS 14 20 30 criteria.

Automatic check against the allowable limit Compares the computed result against the allowable deflection and determines pass/fail.

Automatic structural calculation sheet output Generates an A4 deflection review report that includes the calculation process and results.

A GEN NX API plug-in that automatically generates an initial steel frame model (Node/Column/Beam/Brace/Support) for a two-row, multi-level pipe rack, for structural analysis, from just the bay spacing, story height, and similar input values.

Ultra-fast, parameter-driven modeling (removes repetitive input, adds practical flexibility) Completes a pipe-rack frame in one step from parameter input (bay spacing, story height, etc.) alone, with no complex manual work. Existing model materials and sections can be reused, and new ones can be created and assigned right in the UI, so you can start even from an empty model. Everything from geometry to support conditions is handled in one batch, meaningfully cutting down repetitive modeling time.

Real-time 3D preview with thorough pre-validation (eliminates human error) The 3D preview updates live as parameters change, so you can check the geometry intuitively. Right before committing to the GEN NX model, it validates the expected member count, the ID range to be assigned, and whether there is any conflict with the existing model. The model is only created once validation passes, which eliminates human error.

A plug-in that checks a structure's overturning stability under wind and seismic loads, based on GEN NX analysis results and design conditions entered by the user. It automatically pulls the dead-load support reactions and overturning moments needed for the check straight from GEN NX, so the whole workflow — from reviewing analysis results, through the overturning stability check and governing-condition

Step-by-step guidance for first-time users runs as one continuous process without switching between separate result tables or calculation sheets. An intuitive status bar at the top and pulsing highlights on input fields show the current progress and the next action, so even first-time users can pick up the full workflow quickly.

Automatic linkage of GEN NX results Automatically pulls the dead-load support reactions and the foundation-level overturning moments for the wind and seismic load cases from GEN NX, and derives the resisting moment using the actual lever-arm distance and the self-weight of foundations/walls outside the model.

Overturning stability and governing-condition determination Builds the KDS load combinations per the strength design method or the allowable stress design method, and compares the overturning moment, resisting moment, and safety factor for wind and seismic loads together to determine the governing condition.

Report output with the calculation basis Provides a report-output feature that summarizes the input conditions, GEN NX analysis results, overturning stability check results, and the detailed calculation basis.

A plug-in that collects the full set of declared data (structural system / stories / materials / sections / boundary conditions / loads / analysis options) from the model currently open in GEN NX via the MIDAS API, and automatically generates a "modeling review report" (PDF, Excel) ready to attach directly as an appendix to the structural calculation document.

Shorter model-review time A full review of the model built in GEN NX cuts down the time both senior and junior engineers spend checking the modeling.

Prevents omissions Prevents easy-to-miss omissions in items such as self-weight, boundary conditions, and stiffness factors that are scattered across menus.

Less effort writing review documents Story drift and irregularity checks are compiled automatically as well, reducing the effort of preparing seismic-related review documents.

A single executable (.exe) tool that lets you check off, from a model already analyzed and open in MIDAS GEN NX, analysis images such as the 3D shape, vibration mode shapes, and seismic/wind displacement distribution, along with review graphs such as story drift ratio, story shear, and mass participation ratio, display them on the program screen, and save them — individually or in a batch — as images (JPG/PNG), graphs (PNG), and an Excel (xlsx) file of the review graphs.

Model extraction Read-only — it never changes the model — and automatically recognizes the load names and number of modes from the model.

Turning results into images Automates turning analysis results into images and graphs, cutting report-prep time, with the goal of producing output in the same Excel format already used in practice.

Report review Analysis images and review graphs needed for review-committee submissions and structural review reports can be gathered in a batch with just a few clicks.

Load names Automatically recognizes load names and the number of modes, so it can be reused without reconfiguration even when the model changes.

A plug-in that enables easy, fast apartment analysis modeling from a structural-plan DXF, through automated Wall Mark/ID assignment.

Automatic Wall Mark generation Automatically generates Wall Marks to name walls, which can then be changed to suit the user's preference.

Automatic Wall ID assignment Automatically assigns Wall IDs based on the Wall Marks, to satisfy GEN NX's wall design logic (design per Wall ID).

Reference-story support Multiple stories' DXF files can be considered as needed, and a reference story can be designated so the same Wall Mark/ID assignment is applied identically to other stories.

Convenience features The plug-in lets you check the Material, Properties, and Thickness that will be used before modeling, and handles beam and column elements together as well, making complex modeling easy.

A plug-in that automatically derives wheel reactions from crane specifications and the maximum envelope reaction on the columns supporting the runway beam. With a single button, it generates the load cases needed for the crane's vertical and horizontal loads and applies the computed nodal loads to the GEN NX model in a batch. The entire workflow — from load derivation, through model input, to checking

Step-by-step guidance for first-time users An intuitive status bar at the top and pulsing highlights on input fields show the current progress and the next action, so one continuous process, without switching between a separate calculation tool and GEN NX. Even first-time users can pick up the full workflow quickly.

Maximum envelope reaction for multiple cranes Reflects the simultaneous-operation condition of multiple cranes to compute each crane's wheel reaction, and automatically derives, via influence-line analysis, the maximum envelope reaction for each runway-beam support column.

Batch input of load cases and nodal loads Automatically generates the load cases needed for the crane's vertical and horizontal loads, and applies the computed nodal loads to the selected nodes of the GEN NX model, in a batch, by load case.

Report output with the calculation basis Provides a report-output feature summarizing the basis for deriving the wheel reactions and applied loads from the crane specifications.

A plug-in that checks a structure's buoyancy (flotation) safety at completion and at each construction stage, based on GEN NX analysis results and design conditions entered by the user. It automatically pulls the analysis results needed for the check straight from GEN NX, so the whole workflow — from the stage-by-stage safety factor check, through deriving the control water level, to report output — runs as

one continuous process, without switching between separate result tables or calculation sheets.

Step-by-step guidance for first-time users An intuitive status bar at the top and pulsing highlights on input fields show the current progress and the next action, so even first-time users can pick up the full workflow quickly.

Automatic linkage and cross-validation of GEN NX results Automatically pulls the per-story dead-load axial force and foundation support reactions from GEN NX, then cross-validates the two analysis results to check for missing or inconsistent data.

Stage-by-stage buoyancy safety and control water level Automatically computes the buoyant force, resisting force, and buoyancy safety factor for the completed state and each construction stage, and back-calculates the control water level at each stage needed to satisfy the target safety factor.

Report output with the calculation basis Provides a report-output feature that summarizes the input conditions, cross-validation results, the stage-by-stage buoyancy safety factors, and the basis for the control water level.

A plug-in that lets you select GEN NX wall rebar data (Modify Wall Rebar Data) by Wall Mark or by thickness, then enter the vertical, horizontal, end-confinement, and boundary-element rebar together with cover and thickness for each story range in one step, and export the entered rebar arrangement as a PDF, organized in a table by Wall Mark.

Story repetition Automates the repetitive manual work of applying rebar arrangements across story ranges in a batch.

Wall review Also automatically organizes the model's wall rebar data into a table by mark (story range, rebar, cover, thickness) and exports it as a PDF — output you can use directly for design review and document preparation.

A tool that optimizes the steel sections of a MIDAS GEN NX model.

Reduced material quantity Finds members with spare capacity in GEN NX's design-check results and steps them down, one size at a time, on the standard H-beam size ladder, reducing the total steel quantity.

Automatic member-size decision The plug-in automatically decides the member size closest to a target design ratio specified by the user.

A plug-in that lets you pick a solar structure type (CASE 1–4) in MIDAS GEN NX, and once you enter the dimensions, materials, and sections, automatically generates the nodes, beam elements, and boundary conditions in the GEN NX model.

Model input A plug-in that, once you enter the dimensions, materials, and sections, automatically generates the nodes, beam elements, and boundary conditions in the GEN NX model.

Model computation Once the installation type and key dimensions are entered, the nodes and elements for every component — columns, roof beams, braces, purlins, and so on — are automatically computed and generated, sharply cutting model-build time.

Member computation Compared to manual modeling, the repetitive work of computing node numbers, determining member coordinates, and entering boundary conditions disappears, significantly reducing model-generation time.

Load computation Gravity loads (dead, live, snow) and KDS-based wind loads are automatically computed and assigned, reducing load-input errors and ensuring compliance with the design code.